www.aktunotes.in
www.aktunotes.in

Printed Pages—2 **EEC032**

(Following Paper ID and Roll No. to be filled in your Answer Book)

PAPER ID : 2887 **Roll No.** | | | | | | | | | | |

B. Tech.

(SEM. VIII) THEORY EXAMINATION 2011–12

DIGITAL SYSTEM DESIGN USING VHDL

Time : 3 Hours *Total Marks : 100*

Note :– Attempt **all** questions. Each question carries equal marks.

1. Attempt any **three** parts of the following :– **(20)**
   (a) Define following in context with VHDL :
      (i) Structures
      (ii) Objects
      (iii) Data type.
   (b) Explain sequential statements and process by taking suitable examples.
   (c) Discuss sequential modeling and attributes.
   (d) Explain conditional assignment with the help of an example.
   (e) Explain array loops and assert statements by considering proper example.

2. Attempt any **three** parts of the following : **(20)**
   (a) Discuss basic structure of VHDL.
   (b) Write a VHDL module that describes a 16-bit serial-in, serial-out shift register with input S/ (serial input), EN (Enable) and CK (clock, shift on rising edge) and SO (serial output).

www.aktunotes.in

http://www.UPTUonline.com

www.aktunotes.in
www.aktunotes.in

(c) Write a VHDL description of a SR latch.
  (i) Use a combinational assignment statement
  (ii) Use the characteristic equation
  (iii) Use two logic gates

(d) Write a VHDL code for full subtracter using logic equations.

(e) Explain binding alternatives and genetic parameters.

3. Attempt any **three** parts of the following : **(20)**

(a) Explain concurrent signal assignment with suitable block diagram and VHDL codes.

(b) Discuss guarded signal assignment with suitable VHDL codes.

(c) Explain different sequential statements with suitable block diagram.

(d) Explain declaration for initializing multidimensional arrays, non integer indexing and unconstrained arrays.

(e) Write the instructions used for :
  (i) Opening and closing files
  (ii) File read and write operation.

4. Attempt any **two** parts of the following : **(20)**

(a) Explain inertial and transport delay mechanism with their comparison.

(b) Explain concurrent assignments and sequential placement of transactions.

(c) Explain delta delay and transaction appending rules.

5. Attempt any **two** parts of the following : **(20)**

(a) Discuss issues related to core design test.

(b) Discuss synthesis rules and styles for hardware cores and models.

(c) Explain memory and queue structure. Also, discuss about arithmetic cores.

www.aktunotes.in

http://www.UPTUonline.com